Geometry-Aware Multimodal Fusion for Surface Defect Detection

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Declaration

It is hereby declared that

1. The thesis submitted is our own original work while completing degree at Brac University.

2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.

3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.

4. We have acknowledged all main sources of help.

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Abstract

Multimodal industrial surface defect detection systems integrate RGB and 3D sensor data to achieve high-precision identification of manufacturing defects, yet practical deployment is severely challenged when sensors degrade or fail, resulting in missing or unreliable modality data. Existing approaches typically treat modality absence as a static condition, neglecting to anticipate progressive sensor health deterioration or to leverage geometric priors when visual (RGB) information becomes compromised. This thesis proposes a resilient multimodal framework for industrial defect detection that extends prompt-based learning with two novel contributions: (1) Sensor Sentinel, a predictive health-monitoring module that anticipates sensor degradation before it compromises detection performance, and (2) Geometric Feature Injection, which explicitly encodes curvature and surface normal information to preserve robustness when the RGB modality is unreliable. Built upon cross-modal prompt learning and symmetric contrastive learning foundations, the proposed approach enables adaptive fusion strategies that dynamically respond to real-time sensor conditions. Experimental validation on industrial defect datasets demonstrates that the combination of proactive sensor monitoring and geometry-aware feature representation yields significantly superior detection performance under dynamic modality-missing scenarios compared to conventional reactive missing-modality handling methods. Keywords: industrial defect detection, multimodal learning, sensor health monitoring, missing modalities, geometric feature extraction, prompt learning, RGB-D fusion, predictive maintenance.

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Chapter 1

Introduction

The integration of Artificial Intelligence (AI) into industrial manufacturing has completely revised the standards of quality assurance and operational efficiency. High-precision surface flaw detection is becoming a functional requirement for high-throughput production lines in the era of Industry 4.0, rather than just a competitive advantage. Traditional inspection systems, once reliant on human vigilance and basic template matching, are being replaced by deep-learning-based multimodal architectures. These modern systems aim to replicate human sensory perception by combining visual texture (2D) with structural depth (3D), creating a comprehensive digital representation of physical anomalies. However, these systems face the "unpredictability of the physical world" when they move from the antiseptic laboratory environment to the high-stakes reality of the manufacturing floor. This unpredictability displays as sensor degradation, environmental interference and the dynamic loss of data streams, which can severely hamper a standard AI model's decision-making capabilities.

This chapter introduces a research framework designed to overcome the shortcomings of existing reactive multimodal detection systems. We start by exploring the foundational literature that has shaped the field of Multimodal Industrial Surface Defect Detection (MISDD), specifically focusing on the recent advances of large-scale Vision-Language Models (VLMs) and prompt-based learning. We then establish the justification for this research, proposing that a change from binary data handling to a proactive, "self-aware" architecture is necessary for industrial AI to achieve full resilience. By introducing the concepts of a *Sensor Sentinel* for real-time reliability forecasting and *Geometric Feature Injection* for structural grounding, this introduction outlines a pathway toward a more robust and physically aware inspection paradigm. The following section outlines the specific problem, objectives and methodological scope, providing a roadmap for a framework that ensures inspection integrity remains uncompromised, regardless of facing sensor uncertainty.

1.1 Background

The field of Industrial Surface Defect Detection (ISDD) has experienced a paradigm shift, moving from manual inspection to automated systems driven by Deep Learning (DL). Conventional techniques primarily employed 2D RGB imaging due to its high resolution and ease of integration. However, as manufacturing tolerances have become stricter, the constraints of 2D data have become a hurdle. RGB-based systems are well-known for being highly sensitive to environmental factors such as uneven lighting, shadows and reflections from metallic surfaces, which can lead to high false-positive rates. More importantly, 2D images do not reveal the actual surface features of a product, making it nearly impossible to distinguish between a harmless surface stain and a significant structural dent. This technological gap has catalyzed the adoption of Multimodal Industrial Surface Defect Detection (MISDD), which enhances RGB data with 3D modalities such as point clouds or depth maps. These 3D sensors provide a geometric "ground truth", enabling the system to verify the physical integrity of the surface despite visual noise.

1.1.1 Evolution of Industrial Surface Defect Detection (ISDD)

Industrial Surface Defect Detection (ISDD) has evolved through three distinct technological eras: manual visual inspection, traditional machine vision and deep-learning-based analysis. Initial manual techniques were fraught with subjectivity and fatigue-induced errors. The subsequent era of machine vision introduced high-speed cameras and manually designed features (e.g., Sobel filters, Gabor transforms), yet these systems struggled with the "intra-class variation" of industrial defects. Currently, Deep Learning, particularly Convolutional Neural Networks (CNNs) and Vision Transformers (ViT), allows models to learn hierarchical feature representations. However, as manufacturing progresses to sub-millimetre tolerances, the field acknowledges that 2D RGB data alone fails to capture structural integrity, leading to the need for multimodal fusion to bridge the "texture-structure" gap.

1.1.2 Multimodal Data Fusion and 3D Representation

A comprehensive picture of the physical world can be obtained by combining RGB and 3D data (point clouds, depth maps, or models). The 3D modality acts as a geometric "ground truth" in ISDD that is invariant to lighting conditions. 3D data is crucial for detecting volumetric defects such as dents, bumps, or micro-warping, while RGB is excellent at detecting color-based anomalies like oxidation or stains. Current research focuses on how to best represent this 3D data; while voxelization and multi-view projections were early preferred methods, the direct processing of raw point clouds using Point-based Transformers has become the gold standard for preserving spatial accuracy.

1.1.3 Vision-Language Models (VLMs) and Semantic Anchoring

The rise of models such as CLIP (Contrastive Language-Image Pretraining) has brought a novel idea: semantic anchoring. By integrating visual features with a natural language space, VLMs enable ISDD models to progress from "blind" anomaly detection to "category-aware" evaluation. In this context, a defect refers to a characteristic that semantically relates to terms like "scratched surface" or "deformed edge" instead of just a statistical anomaly. This research makes use of this alignment to ensure that, even when visual modalities are degraded, the remaining sensors can continue to link their features to a known semantic concept of a "defect".

1.1.4 Missing Modality and Prompt Learning

The "modality-missing" problem is one major obstacle to the deployment of multimodal systems. In factories, power surges, network latency, or maintenance can cause sensors to go offline. Recent developments in Deep Correlated Prompting (DCP) [1] and Resilient MISDD-MM [2] have utilized "learnable prompts" to maintain system performance. Instead of retraining the entire model for every possible sensor combination, these systems insert short, learnable parameter sets that serve as "instructions" for the frozen model, effectively "filling in" the information gap left by the missing sensor.

1.1.5 The Proactive Paradigm Shift in Sensor Health

The present state-of-the-art remains reactive, where the loss of modality is modeled as an instantaneous event. However, in the real world, the hardware degrades over time. "Sensor Health Monitoring" is a term that encompasses the monitoring of the signal-to-noise ratio, temporal consistency and calibration of the sensors. This research proposes a paradigm shift where the detection model is "self-aware". This means that the health monitoring module is integrated directly within the model, which can adaptively re-weight its fusion strategy, relying more or less on the sensor based on its health, thereby providing "prevention of silent failures".

1.2 Rational of the Study or Motivation

The disparity between controlled laboratory benchmarks and the chaotic reality of industrial situations serves as the main justification for this research. In research papers, modalities are typically considered "present" or "absent". However, in the real world, the sensors are not binary in terms of health. An RGB camera may have sensor noise from high temperatures, while a laser scanner may encounter "soft failures" such as signal attenuation from lens fogging. Current resilient models are binary; they do not modify their confidence according to the incoming signal's *quality.* Because of this vulnerability, a model may choose to believe a noisy, degraded depth map over a crisp RGB image just because the 3D modality is "present". This research is motivated by the requirement for a "Sensor Sentinel" module—a proactive diagnostic layer that assesses each modality's reliability in real-time and dynamically modifies the fusion weights.

Apart from reliability, there is a fundamental motivation for improving the "structural intelligence" of these models. Conventional Vision Transformers (ViT) are "geometry-blind", meaning that they operate on on 3D point clouds as sequences of coordinates without any inherent understanding of surface continuity or curvature. The shift in surface curvature is a far more reliable signal than raw coordinate values for industrial defects like microcracks or shallow dents. We provide the model a physical inductive bias by explicitly introducing geometric descriptors, such as Gaussian and Mean Curvatures, into the prompting mechanism. This ensures that the structural ”truth” of the item is never disregarded and makes the system much more robust in situations when glare or oil obscures the RGB signal.

1.3 Problem Statement

The main problem addressed by this thesis is the ”Reactive-Blindness” in existing MISDD systems. The deployment of current frameworks in mission-critical manufacturing is limited by two significant flaws. Firstly, they are unable to predict and measure sensor degradation. These models only switch to ”recovery mode” when a sensor is completely offline because they are reactive. As a result, there is a decrease in detection accuracy during the ”soft failure” phase, where the model attempts to process faulty data as if it is correct Second, explicit 3D geometric limitations are frequently overlooked in Transformer designs due to their reliance on end-to-end learnt features. In the absence of explicit structural priors, the model must ”re-learn” the laws of geometry for every new product, which results in poor generalization when dealing with subtle physical anomalies that lack a strong visual contrast.

Furthermore, there is a large disconnect between hardware health monitoring and AI-based inspection. These are two separate pipelines in an actual industrial setup. The AI model continues to produce ”high-confidence” predictions that are fundamentally incorrect if a sensor is somewhat out of calibration. In order to develop a detection system that is both physically aware and truly resilient, The research tackles the urgent need for a unified architecture that combines proactive sensor-health forecasting with geometry-informed feature extraction.

- **Reactive Nature:** Existing MISDD-MM models only compensate for data after it has already been lost.
- **Underutilization of Geometry:** Transformers lack the explicit geometric inductive bias needed for fine-grained structural surface analysis.
- **Disconnected Pipelines:** There is a disconnect between sensor health monitoring and the downstream defect detection logic.

1.4 Objective

The main goal of this research is to architect a next-generation MISDD framework that is based on physical geometry and proactive in its robustness. The research focuses on adding a real-time diagnostic layer and a structural descriptor injection module to the existing prompt-based learning paradigm in order to accomplish this. The goal is to ensure that the detection accuracy stays constant even when sensor quality fluctuates, therefore providing a more reliable solution for automated quality assurance in high-speed industrial lines.

The research specifically aims to go beyond the ”one-size-fits-all” prompting that is present in current literature. We intend to create ”Reliability-Modulated Prompts” that scale their influence based on the predicted health of the sensor. By doing this, the framework will be able to increase its dependence on the explicit geometric features derived from the remaining modalities while also gracefully reducing its reliance on a malfunctioning sensor. This comprehensive strategy ensures that the ”Deep Correlated Prompting” logic is not just correlating abstract features but also guided by the physical and technical state of the inspection environment.

- **Objective 1:** To design a *Sensor Sentinel* module that uses temporal ans signal-quality analysis to predict a reliability score (R) for each modality.
- **Objective 2:** To implement a *Geometric Feature Injection* mechanism that extracts and fuses surface normals and curvatures into the latent prompt space.
- **Objective 3:** Instead of using binary absence, the *Missing-Aware Prompts* (MAP) should be modified to trigger adaptively based on the expected reliability score.

1.5 Methodology in Brief

The methodology is structured into four highly technical phases that are intended to close the gap between signal processing and deep learning:

Phase 1: Baseline Architecture and Semantic Alignment. We implement a dual-branch Vision Transformer (ViT-B/16) using the CLIP-based framework. To align RGB and 3D features, this step entails setting up the ”Cross-Modal Consistency Prompts” (CCP). To map these visual traits into a shared space with antithetical text prompts (e.g., ”defect-free” vs. ”damaged”), we use Symmetric Contrastive Learning (SCL).

Phase 2: Sensor Sentinel Development. We design a Temporal Convolutional Network (TCN) that functions as a proactive diagnostic layer. This module generates a Reliability Tensor $\mathcal{R}$ from a sliding window of sensor signal metrics, such as entropy, peak signal-to-noise ratio (PSNR), and point density. For the next prompt-learning module, this tensor serves as a dynamic gate.

Phase 3: Geometric Induction and Feature Injection. We generate Gaussian or Mean curvature fields and local surface normals using GPU-accelerated Principal Component Analysis (PCA). These are not merely concatenated but are passed through a ”Geometric Projection MLP” to align them with the 768-dimensional CLIP latent space. These features are then inserted as ”Geometric Prompts” into the transformer layers.

Phase 4: Adaptive Multi-Task Optimization. The final phase involves training the learnable prompts using a multi-task loss function: $\mathcal{L} = \alpha\mathcal{L}_{detect} + \beta\mathcal{L}_{align} + \gamma\mathcal{L}_{sentinel}$. This ensures that the model retains cross-modal alignment and sensor-health awareness while learning to identify defects. The MVTec 3D-AD dataset is evaluated using additional ”Industrial Noise Models” (motion blur, point sparsity and Gaussian noise).

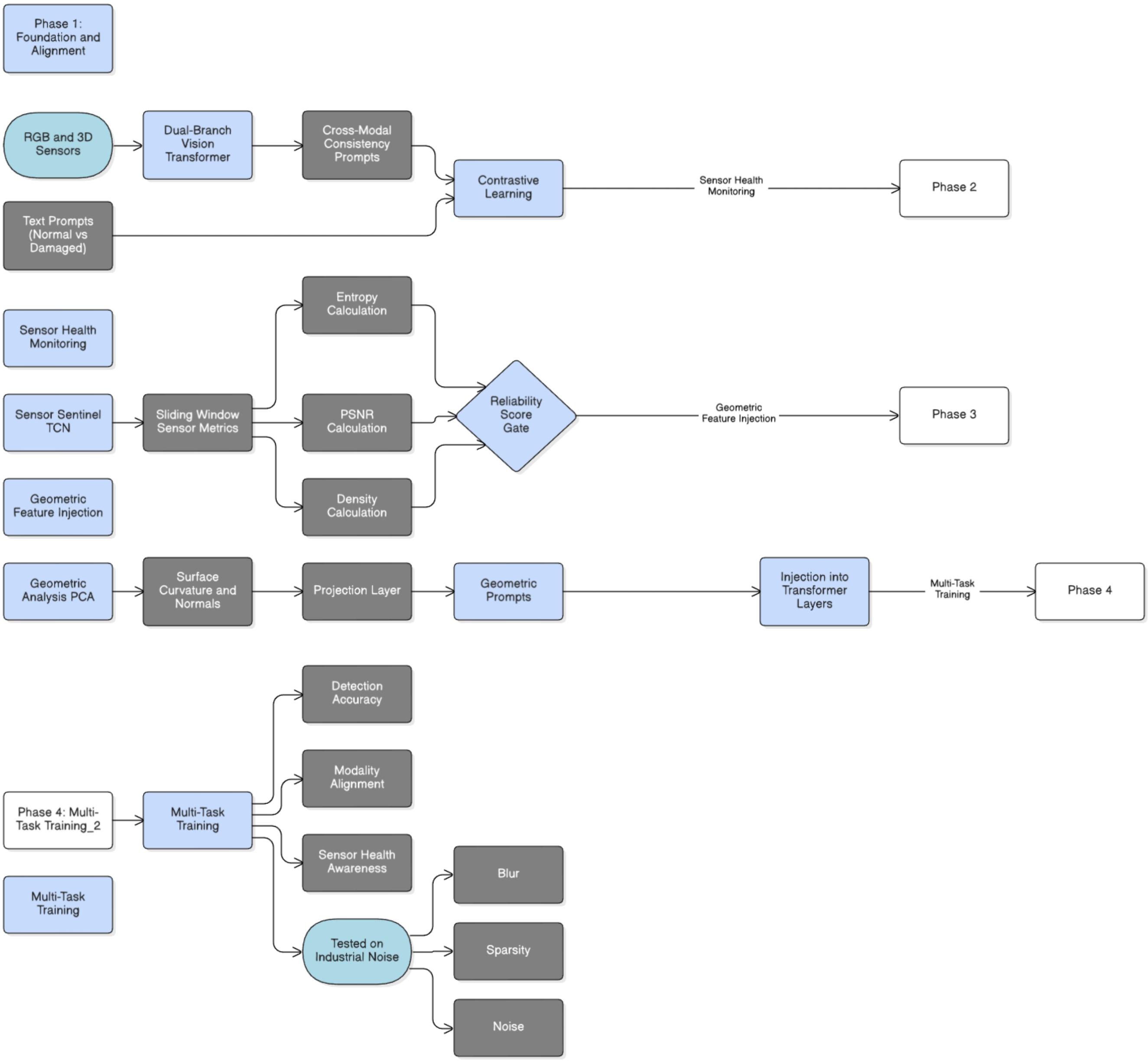

Figure 1.1: Overview of the proposed methodology pipeline .

1.6 Scopes and Challenges

The scope of this research is precisely defined within the framework of industrial surface inspection. We focus on rigid, non-deformable objects that are frequently encountered in production, such as machined parts, electronic casings and engine components. The "3D" modality is explicitly limited to point clouds and depth maps generated by structured light or LiDAR sensors despite the framework's multimodal design. The research does not extend to "soft" materials like textiles or liquids, nor does it address internal defects that require X-ray or ultrasonic penetration.

The main challenges of this research are primarily computational and data-driven. The inference pipeline may be slowed down by the mathematically intensive process of extracting high-resolution geometric features, such as Gaussian curvature. To be practical for a factory floor, this extraction must be optimized with CUDA-based parallel processing to maintain latency below 200 ms. Moreover, publicly available datasets containing "real" sensor failure sequences are hard to come by. To overcome this, we must develop high-fidelity simulation models that replicate dust accumulation, motion blur and sensor drift, so that "Sensor Sentinel" is trained on realistic failure trajectories.

- **Scope Boundary:**Text modality is only utilized as a semantic bridge; it is limited to surface anomalies that can be identified through texture and geometry.
- **Technical Challenge:** Real-time geometric feature extraction (Normals/Curvature) requires optimization and is computationally expensive.
- **Data Constraint:** High-fidelity synthetic degradation simulations are required due to the lack of real-world "sensor failure" datasets.

Chapter 2

Literature Review

Chapter Overview

The Literature Review chapter offers a thorough analysis of the theory and the state of the art in multimodal industrial surface defect detection, including the challenges of sensor reliability and modality incompleteness. The reliance on high-precision automated inspection systems has never been more paramount in the era of smart manufacturing; however, the jump from the controlled environments of the lab to the ever-changing environments of the factory floor poses unprecedented risks with respect to the reliability of the inspection systems. The chapter offers a thorough overview of the evolution of defect detection systems from the initial monocular visual inspection systems to the current systems incorporating surface texture and geometric information. The chapter starts with the establishment of the fundamental architectural preliminaries, including the mechanics of how models process image patches and how language is used to ground the results of the vision systems in semantic space. The chapter also offers an overview of the physics of surface geometry, including how curvature is used to reliably identify structural defects. It also offers a formalization of the problem of missing modality through the lens of sensor degradation. By bringing together the diverse domains of computer vision, natural language processing, and physics, the chapter offers a rigorous conceptual basis while highlighting the research gaps with respect to resilient fusion systems, which is the focus of the current work.

2.1 Preliminaries

The chapter starts with the establishment of the fundamental concepts, mathematical notation, and architecture used in the current multimodal industrial surface defect detection systems. It offers a formal definition of the key concepts, the transformer architectures used to power the current vision systems, the vision language alignment, and the problem of missing modality in the context of resilient systems.

2.1.1 Mathematical Notation

The following table summarizes the main mathematical notation used throughout this thesis for consistency and clarity.

Symbol	Description
x, y, η	Scalars
$\mathbf{x}, \mathbf{f}$	Vectors
$\mathbf{X}, \mathbf{W}$	Matrices
$\mathcal{D}, \mathcal{G}$	Sets
N	Number of samples
C	Number of channels
H, W	Height and width of an image
d	Dimension of the embedding/feature space
L	Sequence length
l	Prompt length
M	Total number of modalities
$m \in \{m_1, m_2, m_3\}$	Modality index (RGB, 3D point cloud, text)
$\mathbf{D}^m$	Raw input data from modality m
$\mathbf{F}^m$	Extracted feature representation of modality m
$\otimes$	Element-wise (Hadamard) product
$\circ$	Function composition
$\|\cdot\|$	Norm operator
$\langle \cdot, \cdot \rangle$	Inner (dot) product
$\mathrm{Concat}(\cdot, \cdot)$	Concatenation along a dimension
$\eta \in [0, 1]$	Missing data rate
$\mathbf{M}_i = (M_i^{m_1}, M_i^{m_2}) \in \{0, 1\}^2$	Binary missingness mask for sample i
$\tilde{\mathbf{D}}^m = \mathbf{D}^m \otimes M^m$	Simulated missing data for modality m

Table 2.1: Main mathematical notation.

2.1.2 2.1.2 Vision Transformer (ViT) Architecture

The Vision Transformer [3], revolutionized image understanding by representing images as sequences of patches instead of a grid of pixels. Unlike traditional CNNs that apply local context modeling with self-attention only in the final layer, ViT uses self-attention from the first layer. Given an input image $\mathbf{X} \in \mathbb{R}^{H \times W \times C}$, we divide it into $P = (H/p) \times (W/p)$ patches, where each patch is a $p \times p$ sub-grid. Each flattened patch $\mathbf{x}_i \in \mathbb{R}^{p^2 \cdot C}$ is then projected to a space with dimension d by $\mathbf{E} \in \mathbb{R}^{d \times (p^2 \cdot C)}$. We then append a class token $\mathbf{z}_{cls}$ and positional embeddings to form a sequence $\mathbf{Z}^{(0)} = [\mathbf{z}_{cls}^{(0)}; \mathbf{z}_1^{(0)}; \ldots; \mathbf{z}_P^{(0)}] \in \mathbb{R}^{(P+1) \times d}$.

The input sequence is then processed by N layers, where each layer consists of a Multi-Head Self-Attention (MSA) and a Feed-Forward Network (FFN):

$$\mathbf{Z}'^{(\ell)} = \mathrm{MSA}(\mathrm{LN}(\mathbf{Z}^{(\ell-1)})) + \mathbf{Z}^{(\ell-1)} \tag{2.1}$$

$$\mathbf{Z}^{(\ell)} = \mathrm{FFN}(\mathrm{LN}(\mathbf{Z}'^{(\ell)})) + \mathbf{Z}'^{(\ell)} \tag{2.2}$$

Here, FFN is defined as $\text{FFN}(x) = \mathbf{W}_2 \cdot \text{GELU}(\mathbf{W}_1 x + \mathbf{b}_1) + \mathbf{b}_2$. Standard self-attention is $\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}((\mathbf{Q}\mathbf{K}^T)/\sqrt{d_k})\mathbf{V}$. $\text{Attention}(\mathbf{V}, \mathbf{V}, \mathbf{V}) = \text{softmax}((\mathbf{V}\mathbf{V}^T)/\sqrt{d_k})\mathbf{V}$, is a consistent self-attention variation that [2] present precise localization. This promotes local feature consistency, which is crucial for spotting tiny industrial anomalies.

2.1.3 Vision-Language Models: CLIP

Contrastive Language-Image Pre-training (CLIP), which uses extensive contrastive training to align textual and visual representations in a single space [4]. Two encoders are used in the model: a text encoder f_T and an image encoder f_I. For an image $\mathbf{I}$ and text $\mathbf{T}$, encoders produce normalized embeddings $\mathbf{v}_I = f_I(\mathbf{I})/\|\mathbf{f}_I(\mathbf{I})\|$ and $\mathbf{v}_T = f_T(\mathbf{T})/\|\mathbf{f}_T(\mathbf{T})\|$. In a batch of size B, similarity is computed as $S_{ij} = \langle \mathbf{v}_I^{(i)}, \mathbf{v}_T^{(j)} \rangle / \tau$, where τ is a learnable temperature. The model optimizes a symmetric contrastive loss:

$$\mathcal{L}_{\text{CLIP}} = \frac{1}{2B} \sum_{i=1}^{B} \left[-\log \frac{\exp(S_{ii})}{\sum_{j=1}^{B} \exp(S_{ij})} - \log \frac{\exp(S_{ii})}{\sum_{j=1}^{B} \exp(S_{ji})} \right] \tag{2.3}$$

By comparing image embeddings to class-specific prompts (such as "a photo of a [CLASS]"), CLIP [4] makes zero-shot categorization possible. Through terms like "surface scratch", this gives MISDD semantic basis, enabling adaption to new items without requiring significant re-labeling. Nevertheless, CLIP lacks the feature of modality handling and 3D processing. In order to bridge the gap between semantic comprehension and geometric inspection, our research expands this dual-stream architecture to include 3D geometric information through a robust prompting mechanism that keeps alignment even in the absence or degradation of visual cues.

2.1.4 Prompt Learning in Vision-Language Models

Prompt learning is a computationally efficient adaptation strategy for frozen models like CLIP for downstream tasks. Instead of fine-tuning, it involves the use of a few vectors for adaptation. Text prompts, as popularized by CoOp [5], replace the context vectors with learnable vectors $\{\mathbf{p}_i\}_{i=1}^{l}$ from the text embedding space. For image prompts, tokens are prepended to the image patch sequence, while Deep Prompting, a variant known as MaPLe, involves the addition of parameters to multiple intermediate layers for adaptation. The dynamic prompt learning variants, like CoCoOp [6] use a combination of input features for prompt generation using a few neural network layers.

Here, the prompt parameters $\{P\}$ are the only parameters optimized, with the backbone network parameters θ_{frozen} remaining frozen. This is done by optimizing the following loss function:

$$\min_{P} \mathcal{L}(\text{Model}(X; \theta_{frozen}, P), Y)$$

This is especially beneficial for the industrial environment with limited defect data, as only $0.1 - 2\%$ of the total parameters need to be optimized. The "Reliability-Modulated Prompts" variant, as presented in the following sections, overcomes the limitations of the standard prompts by dynamically scaling the prompts according to the sensor reliability in real time. If the 3D sensor is faulty, the geometric prompts will be down-weighted, with the RGB semantic prompts taking over. This balances the benefits of the robustness of the backbone network with the flexibility required for the changing conditions found in modern automated manufacturing environments.

2.1.5 Problem Formalization: Missing Modality in MISDD

The missing modality problem in MISDD [2] needs to be normalized to establish the connection between theory and practice. A multimodal dataset $\mathcal{D} = \{(\mathbf{x}_i^{m_1}, \mathbf{x}_i^{m_2}, y_i)\}_{i=1}^{N}$ is used to represent the data. Specifically, $\mathbf{x}_i^{m_1} \in \mathbb{R}^{H \times W \times 3}$ represents the RGB images and $\mathbf{x}_i^{m_2}$ represents the 3D data. For localization tasks, $\mathbf{y}_i^{pixel} \in \{0,1\}^{H \times W}$ represents the ground truth masks. Due to the limitations of real-world sensors, there are hard failures and soft failures. We describe three cases: missing both (randomized), missing m_1 (RGB lost), and missing m_2 (3D lost). The percentage of incomplete samples is quantified by the missing rate $\eta \in [0,1]$.

Each sample is represented by a binary mask $\mathbf{M}_i = (M_i^{m_1}, M_i^{m_2}) \in \{0,1\}^2$, with missing modalities represented by zeros: $\tilde{\mathbf{x}}_i^m = \mathbf{x}_i^m \otimes M_i^m$. The key problem is to design a model $f(\tilde{\mathbf{x}}^{m_1}, \tilde{\mathbf{x}}^{m_2}; \theta)$ to effectively cope with the presence of both binary absence and continuous quality degradation. The evaluation metrics are Image AUROC (I-AUROC) for detection and Pixel AUROC (P-AUROC), AUPRO for segmentation. This problem formalization allows us to design our "Sensor Sentinel" to adapt from high-confidence multimodal fusion to robust single-modal performance based on the reliability of the sensor $R(t)$, which may fluctuate. We may systematically assess how well the architecture preserves its diagnostic accuracy even when industrial noise or sensor failure compromises the integrity of the input stream by defining the problem space.

2.1.6 3D Data Representations

Industrial systems employ different 3D representation formats to store geometric information of surfaces. A point cloud is an unordered set of coordinates $\mathcal{P} = \{\mathbf{p}_i\}_{i=1}^{P}$ where each coordinate $\mathbf{p}_i = (x_i, y_i, z_i)$. These 3D representations are good for raw geometry with arbitrary topology but are irregular for standard transformer models. Depth maps are 2D grids $\mathbf{D} \in \mathbb{R}^{H \times W}$ that store information about depth. These are good for alignment with RGB images since camera intrinsics are known. A point $\mathbf{p}_i$ can be reconstructed from its depth value D_{ij} as follows:

$$\mathbf{p}_i = \left(\frac{(u_i - c_x) \cdot D_{ij}}{f_x}, \frac{(v_i - c_y) \cdot D_{ij}}{f_y}, D_{ij} \right) \tag{2.4}$$

Surface normal maps $\mathbf{N} \in \mathbb{R}^{H \times W \times 3}$ represent local orientation. Normals can be computed from depth gradients:

$$\mathbf{N}_{ij} = \frac{(\nabla_x D, \nabla_y D, -1)}{\|(\nabla_x D, \nabla_y D, -1)\|} \tag{2.5}$$

While depth maps are effective for backbone networks, they may have "holes" on surfaces of industrial objects. The proposed framework overcomes this challenge by employing the grid-based efficiency of depth maps for primary feature extraction while simultaneously computing explicit geometric features. Therefore, it ensures that the "truth" of an object's geometry is maintained even if its texture is occluded.

2.1.7 Geometric Feature Descriptors

Quantitative geometric features are provided in an explicit manner to measure the properties of the surface, which are important in defect detection. For example, the surface normals $\mathbf{n}_i$ are estimated using PCA on a local neighborhood $\mathcal{N}_k(i)$. The covariance matrix $\mathbf{C}$ is defined as follows:

$$\mathbf{C} = \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} (\mathbf{p}_j - \bar{\mathbf{p}})(\mathbf{p}_j - \bar{\mathbf{p}})^T \tag{2.6}$$

The principal curvatures κ_1 and κ_2 are quantitative features of the bending of the surface, which are the eigenvalues of the shape operator. Using the principal curvatures, the Gaussian curvature $K = \kappa_1 \cdot \kappa_2$ and Mean curvature $H = \frac{(\kappa_1 + \kappa_2)}{2}$ can be calculated.

The Gaussian curvature is a powerful tool to identify the intrinsic properties of a shape: K is positive in peak or valley features, negative in saddle points, and zero in flat points on a surface. These features are based on physical properties of a defect, where a dent has a specific signature in the Gaussian curvature field. By explicitly computing the features and incorporating them into the model, it has a structural inductive bias. It ensures that the model is not relying on visual features to detect defects on a metal or specular surface, which would fail to distinguish a stain on a surface (2D defect) from a dent (3D defect) due to varying lighting conditions.

2.1.8 Sensor Degradation Modeling

Signal quality loss in industrial sensors can be caused by noise, blur or measurement error. We use the Signal-to-Noise Ratio (SNR) and Contrast Ratio (CR) to measure degradation:

$$\text{SNR} = 10 \log_{10} \frac{\sigma^2_{\text{image}}}{\sigma^2_{\text{noise}}} \tag{2.7}$$

$$\text{CR} = \frac{L_{\max} - L_{\min}}{L_{\max} + L_{\min}} \tag{2.8}$$

where brightness is indicated by L. Gradient magnitude at edge pixels is used to quantify sharpness. Random distortion, motion blur kernels and additive Gaussian noise are typical patterns.

To enable adaptive response, we define a composite sensor reliability score $R(t) \in [0, 1]$:

$$R(t) = w_1 \cdot \text{SNR}_{\text{norm}}(t) + w_2 \cdot \text{CR}_{\text{norm}}(t) + w_3 \cdot \text{Sharpness}_{\text{norm}}(t) \tag{2.9}$$

where $\sum w_i = 1$. The "Sensor Sentinel" can identify approaching failures due to this score. In order to move trust away from a deteriorating sensor, the system constantly modifies its internal weighting. This thesis's primary contribution is its proactive strategy, which shifts the state of the art from reactive "missing-aware" prompting to a proactive, health-aware framework. By prioritizing the most trusted data stream that is accessible at any given point in the production cycle, the architecture can retain the best possible inspection accuracy while simultaneously triggering maintenance alerts by continuously monitoring $R(t)$.

2.2 Review of Existing Research

Large Multimodal Models (LLMs) have drastically transformed the way AI systems work with various types of data, such as image, text, and audio, especially benefiting educational applications and content creation. Adversarial attacks create a serious security risk for these models. It takes advantage of the weakness in how the models learn and connect data across different modalities, particularly while working with unpaired data. Because in the case of unpaired data, biases and vulnerabilities are harder to control. The research analyze these security threats and proposes a solution to make multimodal systems safer and more reliable, highlighting the need to establish better frameworks that can deal with various unpaired data while maintaining steady performance [7].

Moreover, recent work like Drive Anywhere leverages multimodal foundation models for end-to-end autonomous driving, illustrating the potential of large-scale vision-language models to generalize in open-set and out-of-distribution situations. They introduced patch-wise feature extraction to maintain spatial reasoning, and their method uses language-augmented latent space simulations for data augmentation and policy debugging. Although the approach proves effective in enhancing the robustness of autonomous driving, it still relies on paired multimodal attributes and particular task supervision, which limits its ability to fully utilize vast amounts of unpaired data. This underlines the need for more coherent and data-efficient frameworks that can fuse and align unpaired image-text representations, which the thesis aims to do [8].

Using a collaborative prompting approach, the MFLCP architecture addresses incomplete modalities in multimodal unified learning. The system enables interaction across available and missing modalities by using modal projection-aware prompts, which is crucial to understanding from partially paired data. The design provides a basis for sensor-aware defect detection by using specialized missing-type and fusion prompts to preserve resilience when certain sensors fail. By leaving big pretrained models frozen and just updating learnable prompts, this lightweight method maximizes communication. Experiments on the UPMC Food-101 dataset show that the model outperforms the PMF baseline of 83.87%, with an accuracy as high as 87.11% even with a 90% missing modality rate [9].

Furthermore, a recent research proposed a VAE-driven multimodal deep learning system that combines structural echocardiographic data and chest X-rays for the early diagnosis of cardiac conditions like SLVH and DLV. In comparison to VisualBERT, CLIP, and other baselines, their model achieves better diagnostic accuracy by combining EfficientNetB3 with attention mechanisms (SE-Block and CBAM), transformer encoders for structural input, and VAEs for latent space fusion. In the medical domain, it is highly effective; however, the technique based on paired multimodal datasets and curated health records, underlining the larger research gap of generalizable methods for learning from unpaired data [10].

According to recent multimodal learning studies, when data is inconsistent or insufficient, performance is not necessarily improved by adding more modalities. Research on incomplete brain tumour segmentation demonstrates that, if modality robustness is not taken into account, the naive fusion of different modalities might add noise and lower representation quality. By applying region-level calibration and learning modality reliability, the PNDC approach enhances performance and generates more reliable fused representations in missing modality situations. According to experimental findings, reliability-aware fusion techniques can achieve better segmentation accuracy with partial modality settings than previous models like RFNet and mmFormer [11].

With the objective to enable a high-resolution and low-noise electrophysiological evaluation of 3D organoids, this recent research proposes a CMOS-based multimodal electrode array that can record, stimulate, and perform impedance spectroscopy across seven modalities. This research prioritizes sensor-level fusion over representation learning or cross-modal alignment. It also stresses hardware scalability and multimodal signal integrity. This reveals a gap in developing a learning framework that can efficiently describe unpaired and diverse multimodal data beyond hardware-level fusion and also highlights the importance of scalable multimodal unification [12].

The Multi-omics Trustworthy Integration Framework (MoTIF), the framework proposed by a recent research study, uses dynamic feature and modality selection along with estimation of uncertainty to improve interpretability and reliability in multimodal learning. The method combines early and late fusion techniques to dynamically allocate confidence to each modality, and it improves both prediction accuracy and reliability. However, scaling to unpaired or diverse modalities is not addressed by MoTIF; it depends mainly on paired multimodal omics data. Though this study points out a research gap in applying such dependable learning methods to unpaired multimodal datasets for wider generalization and underlines the importance of meaningful and dynamic alignment across modalities [13].

For multimodal knowledge graph completion, this research suggests a CLIP-enhanced Hyper-node Relational graph Attention Network (HRGAT), which incorporates textual, visual, and numerical information via low-rank multimodal fusion. The model effectively aligns image-text modalities and uses relation-specific attention to enhance forecasting missing links by using pre-trained CLIP embeddings. The approach shows increased ranking efficiency, particularly for head entities. This research leaves an opportunity to work on dynamic multimodal alignment and robust integration across various unstructured knowledge sources to increase scalability to unpaired data [14].

To address the problem of description generation using unaligned data, a prompt-based framework for Unpaired Image Captioning (UIC) is presented in this research, which utilizes Vision-Language Pre-Trained Models (VL-PTMs) that include a semantic prompt for aggregating cross-domain cues with visual features and a metric prompt for iteratively improving the system using high-quality pseudo-samples. This adversarial learning-based approach is found to improve representation learning using unpaired data, as required for robust multimodal fusion in data-constrained settings. Using a prompt-based approach rather than a traditional reinforcement learning-based approach, the framework is found to outperform existing methods on standard benchmarks. In particular, using the MSCOCO dataset, the effectiveness of the presented approach is demonstrated by obtaining a BLEU-4 score of 18.2 and a CIDEr score of 62.1, outperforming existing methods for unpaired image captioning [15].

Another research suggests a parameter-efficient multimodal transfer learning model for Remote Sensing Visual Question Answering (RSVQA). The approach consists of a multimodal fusion module with self-attention and cross-attention mechanisms, a visual encoder initialized with weights pretrained on large-scale RSIs and a lightweight adaptor. Though, this approach successfully combines textual and visual information to achieve state-of-the-art performance on RSQV datasets, it still mostly relies on paired image-text data for training and uses different encoders for each modality. This research proposes a scope to develop a unified, scalable multimodal learning framework that can utilize unpaired data and mitigate augmentation-induced biases [16].

The study presents Multimodal Aligned Conceptual Knowledge (MACK), a knowledge-based framework for unpaired image matching that uses self-supervised bidirectional cycle-consistency to align words with prototypical vision region representation. The architecture complements cutting-edge models like CLIP and ALBEF by utilizing region-word similarity pooling and re-ranking techniques. Experimental results on MSCOCO and Flickr30k demonstrate notable gains in cross-dataset and zero-shot matching tasks. However, the work highlights various possibilities to improve multimodal alignment by pointing out weaknesses, including dependency on a non-SOTA detector (BUTD), limitation to nouns and adjectives, and absence of dynamic knowledge updates [17].

For the purpose of analyzing resilience against noisy or misaligned image-text pairs, this research links nonlinear contrastive loss to singular value decomposition (SVD), in order to provide a theoretical foundation for multimodal contrastive learning (MMCL). With the vision to improve representation recovery over unimodal self-supervised methods, this study suggest a semi-supervised extension of CLIP which enables the use of unpaired datasets via spectral graph partitioning and improved contrastive loss. The benefits of using unpaired data for improved feature alignment are represented by numerical tests on both synthetic and actual datasets. The work is limited to linear contexts and proof-of-concept trials, which highlights a research gap in scalable, non-linear multimodal architectures: developing reliable, unpaired multimodal representational learning techniques [18].

Attempting to capture cross-modal knowledge without spatial alignment, recent research presents a class-specific affinity-guided fully convolutional network (CSA-FCN) for unpaired multimodal image segmentation. It combines shared convolutional layers with modality-specific batch normalization and class-specific affinity matrices. This method notably outperformed UMMKD and single-modality baselines, achieving a standard Dice score of 87.65% (LGE) and 89.95% (bSSFP) on the MS-CMRSeg 2019 dataset. Similarly, it achieved a smaller Hausdorff distance across modalities on MM-WHS, reaching up to 96,07% Dice (MRI). Despite all of this progress, the framework only supports 2D feature maps and has no deeper interpretability for 3D extensions, leaving a research gap [19].

The existing multimodal AI systems mostly rely on paired datasets; therefore, the research offers a framework, MUM (Multi-Uni-Modal), that can handle both paired and unpaired data by using shared transformer encoders and task-specific heads with independent optimizers. Various cardiovascular inputs (ECG, echocardiography, and phonocardiograms) are transformed into unified embeddings by the architecture, which leads to 16.9% improvement in the heart murmur detection task. Large-scale experiments show that MUM improves task performance by 2.7-4% while reducing 53.5% of learnable parameters. The research gap indicates a need to integrate continuous learning for dynamic clinical contexts with evolving pathologies and expand beyond cardiovascular diagnosis to other medical domains [20].

To overcome paired dataset dependency in multimodal medical AI, CLIP-IT pseudo-pairs unpaired TCGA pathology reports with histological images using CONCH-based CLIP retrieval. Knowledge distillation and LoRA- adapted encoders are used for text-to-vision translation with late fusion. The model outperforms all multimodal baselines in most cases, achieving 1.5-4% accuracy improvements on PCAM, BACH, and CRC datasets with little computational expense (0.006% FLOPs increase). The vulnerability towards external text quality and relevance (lower CRC performance with sparse reports), the lack of coordination with other modalities, and the limited exploration beyond classification tasks are research gaps that are specifically related to improving multimodal representation learning for unpaired data by enhancing semantic alignment [21].

the Strong and Weak Prompt Engineering (SWPE) framework to enhance cross-modal retrieval by improving both global and fine-grained semantic prompts. The model employs a dedicated generation module along with a fine-tuning framework based on the transformer architecture, enabling the incorporation of adaptive prompts along with high-level features in an effective manner, especially in the presence of sophisticated background noise. Moreover, an adaptive hard sample elimination module is employed in the model to optimize the training process by weighing the negative pairs of samples in an adaptive manner. This model is highly applicable in robust industrial sensing, especially in situations where accurate feature matching is required in the presence of interference. The model has demonstrated state-of-the-art performance on the RSICD dataset [22].

The core innovation of the paper ”L-MCAT” is the Unpaired Multimodal Attention Alignment (U-mAA), which is specifically exciting as it directly addresses my objective of designing training strategies for truly unpaired data. It demonstrates how models can acquire cross-modal connections without explicit spatial correspondence or labels by integrating a contrastive self-supervised mechanism into the transformer attention. Additionally, the paper’s focus on lightweight architectural modifications, like Modality-Spectral Adapters, align perfectly with the thesis aim to improve computational viability and decrease hurdles, making sophisticated multimodal methodologies more accessible for research and deployment [23].

In order to address the data shortage, "Multimodal Data Augmentation for Image Captioning Using Diffusion Models" offers a crucial perspective. Their approach of using Stable Diffusion to produce excellent artificial image-caption pairs from text alone strikes me as especially perceptive. It shows that models trained on fully synthetic datasets can outperform traditional unpaired image captioning methods, achieving significant improvements, particularly in few-shot scenarios with limited labeled data. Our architectural enhancements focused at computational efficiency benefit greatly from the study's multimodal augmentation approach, which combines synthetic image generation with caption expansion through paraphrasing. Our strategy for sturdy training frameworks is driven by their results that suitable data filtering based on quality criteria (CLIPScore) can improve both training efficiency and performance [24].

Whenever ultrasound paired with microflow imaging samples are unavailable, GMRLNet, a graph-based manifold regularization learning framework addressing incomplete multimodal datasets in medical imaging. Its framework introduces a Graph Convolutional Network-based Shared and Specific Transfer Network (GSSTN) that separates modal inputs into interpretable shared and particular feature spaces. It concurrently captures global data distributions, local inter-sample relations and sample-level features. Unlike GAN-based incomplete learning methods that generate pseudo images, GMRLNet achieves improved performance (0.913 AUC) even with 80 percent missing modality [25].

By merging structured tabular clinical records with unstructured discharge summaries, the paper Frattallone-Llado addresses temporal event extraction from healthcare data, highlighting the major benefits of multimodal approaches over text-only methods. Their annotation methodology supports our goal for enhanced multimodal representation learning by demonstrating that multimodal access reduces temporal uncertainty by 42%, 36%, and 13% for lower limits, upper bounds, and event durations, respectively. Importantly, their multimodal BERT architecture incorporates structured event data through multi-head attention mechanisms, performs 10% better than unimodal BERT and 80% better in F1 scores for bound predictions. Eventually, this validates our target of making architectural changes to foster cross-modal understanding. We emphasize on learning from truly unpaired scenarios, and the framework's success with unpaired temporal relations where textual events lack explicit structured data timestamps is consistent with this goal [26].

By identifying and alleviating two types of data inconsistency: Modality-Modality (M-M) inconsistency, where information exists exclusively in one modality, and Modality-Label (M-L) inconsistency, where labels explain only one modality's content, a crucial problem in multimodal learning. According to their validation, inconsistency occurrences are present in 63-76% of multimodal datasets, directly impacting retrieval performance. With a retrieval accuracy boost of up to 25.13%, the IA-CMR framework's approach of disentangling modal representations into modality-common and modality-unique hash codes through contrastive learning is in line with our architectural enhancement objectives [27].

2.3 Summary of Key Findings

According to the literature study, multimodal learning has advanced significantly, but the majority of existing models rely largely on large-scale paired or weakly paired datasets, restricting their scalability and performance in real-world scenarios where such paired data is not available. In order to deal with unpaired data, some recent works have introduced methods like cycle-consistency, pseudo-pairing, and self-supervised learning to handle unpaired data. These approaches are often task-specific or computationally expensive.

Another recurring issue is modality inconsistency, which occurs when information is only available in one modality, or label inconsistency, which occurs when labels only describe a part of the data. Numerous studies also highlight how crucial computational efficiency is, as larger models struggle with high resource requirements. Lightweight architectures and knowledge distillation are gaining traction. Models that can form unstructured and unpaired data and manage real-world inconsistencies and generalize well are becoming more and more essential in a variety of fields, including knowledge of graphs, autonomous driving, and healthcare.

The main takeaway is that future research should concentrate on developing frameworks for learning from unpaired multimodal data that are more reliable, effective, and broadly applicable, including improvement in semantic alignment without relying on paired supervision, adopting lightweight models, and designing adaptable systems that can adapt to various data types and domains.

2.4 Work Management Plan

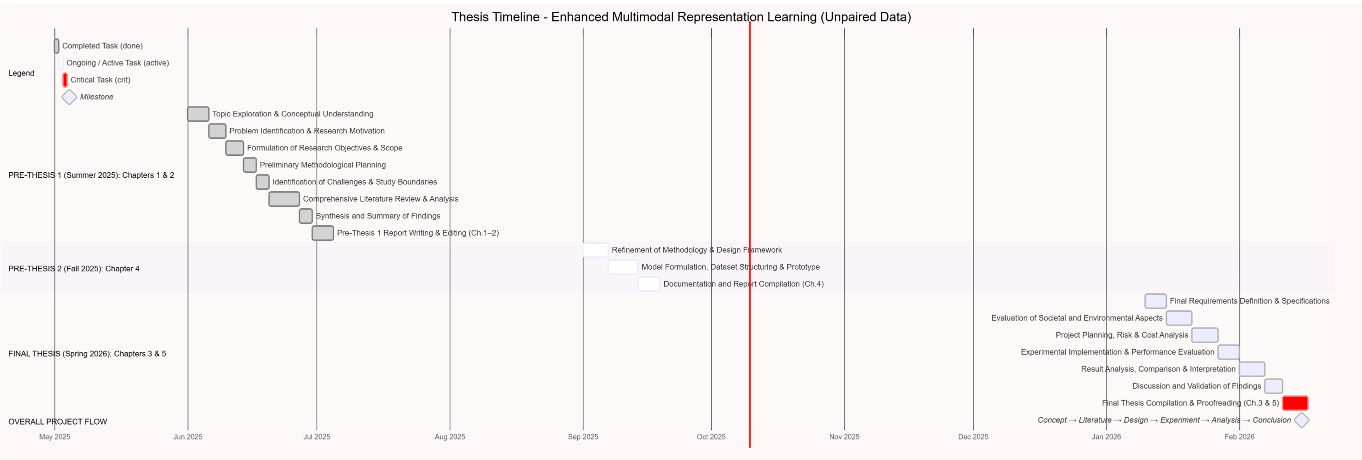

Figure 2.1: Gantt chart representing the structural development of the research paper on Geometry-Aware Multimodal Fusion for Surface Defect Detection across Pre-Thesis 1, Pre-Thesis 2, and Final Thesis phases, it illustrates the thesis timetable.

2.5 Conclusion

The first chapter demonstrates that Multimodal Industrial Surface Defect Detection (MISDD) requires a stronger resilient approach which needs proactive methods. The combination of RGB and 3D data has improved quality assurance standards in Industry 4.0 however the physical world unpredictability creates a major challenge because it causes sensor deterioration and results in unpredictable data loss. The traditional reactive systems handle absence of modality as permanent state which fails to address the increasing sensor health problems that arise over time.

This study presents the MISDD-MM framework which establishes a new approach through its automated data fusion system that operates with intelligent architectural design. The proposed methodology uses a ”Sensor Sentinel” system which provides ongoing reliability prediction together with the ”Geometric Feature Injection” method that establishes structural support during times when visual data becomes unusable to protect inspection processes from environmental disturbances. The framework achieves strong cross-modal comprehension ability through Vision-Language Models (VLMs) which use semantic anchoring and adaptive prompt-based learning to handle situations with extreme missing-modality. The chapter establishes complete system requirements for building a durable inspection system which uses various methods to assess its reliable performance. The following chapters will present the complete system design together with testing methods needed to achieve operational success on the manufacturing floor after laboratory testing has been completed.

Chapter 4

Proposed Methodology

4.0.1 Notation

The following symbols are used throughout this work:

Symbol	Meaning
P_{cons}, P_{spec}, P_{miss}	The three prompt types from CPL (baseline)
P_{geo}	Geometric prompt (our contribution)
R_m	Modality reliability score $\in [0, 1]$ (our contribution)
F_m	Visual feature embedding from modality m
T_n, T_{an}	Normal / abnormal text embeddings
η	Missing rate during training

Table 4.1: List of main symbols and their meanings.

4.1 Design Process or Methodology Overview

The proposed methodology is intended to address the problem of Multimodal Industrial Surface Defect Detection (MISDD) with uncertain sensor availability [2]. The methodology is built to ensure that its detection capability is not compromised, even in situations where one or more modalities, such as RGB or 3D, are dynamically missing during training or inference. This work focuses on sample-specific and dynamic modality absence due to sensor failures, environmental limits, or adaptive acquisition strategies, in contrast to current modality-incomplete approaches that presume static data incompleteness.

The overall design process is divided into three steps, all of which are interconnected with respect to addressing the problem:

1. **Missing Modality (MM) Configuration:** This module applies modality-specific missing masks at the input level or feature level to simulate dynamic sensor availability. Three different data patterns are produced by the configuration: RGB-only, 3D-only and modality-complete. The probability of absence for each modality is controlled by the missing rate η.

2. **Cross-Modal Prompt Learning (CPL):** Instead of training separate models for each input pattern, this step is focused on introducing learnable prompt vectors that are injected into transformer layers. The prompt vectors allow the model to:

   (a) establish cross-modal information consistency with Cross-Modal Consistency Prompt (CCP),

   (b) learn to adapt to different input patterns with a Modality-Specific Prompt (MSP),

   (c) Compensate for missing information using Missing-Aware Prompt (MAP).

3. **Symmetric Contrastive Learning (SCL):** This stage is intended to leverage text modality as a semantic bridge to combine RGB and 3D features. SCL aligns visual representations with textual defect descriptions by using triple-modal contrastive pretraining and antithetical text prompts such as normal and abnormal states. This allows for robust anomaly identification regardless of the type of sensor data that is available.

The main goal is to learn a unified feature space that allows modality-specific encoders to function seamlessly with incomplete inputs while ensuring high defect localization accuracy.

4.1.1 Missing Modality Configuration

Three different modalities are taken into consideration in this work: text, 3D, and RGB. However, sensor availability is unpredictable and only RGB and 3D modalities gathered from sensors are susceptible to simulated absence. RGB-only, 3D-only, and modality-complete are the three separate data patterns that come from the emergence of three different type of missing cases such as RGB missing, 3D missing and both missing.

The absence of modality can happen at the **input level** or **feature level** (pseudomissing). Feature-level missing is intended for memory bank-based methods, like M3DM [28] and Shape-Guided [29]. In order to ensure fair comparison among different baseline architectures, the modality missing type and pattern are decided solely based on the sample, regardless of where the missing operation is conducted. In addition, the data pattern is defined by an indicating matrix $\mathbf{M} \in \mathbb{R}^{C\times H\times W}$, C, H, and W are the height and width of the input or feature maps. In particular, $\mathbf{M}$ defined by following these rules:

$$\mathbf{M}_i^{\text{rgb}}, \mathbf{M}_i^{\text{3D}} = \begin{cases} \langle 0, 1\rangle, & \text{if } \mathbf{D}_i^{\text{rgb}} \text{ is missing} \\ \langle 1, 0\rangle, & \text{if } \mathbf{D}_i^{\text{3D}} \text{ is missing} \\ \langle 1, 1\rangle, & \text{otherwise.} \end{cases} \tag{4.1}$$

Input-Level Missing

The simulation method employed in this work is modality missing at the input level, as this method is more likely to simulate real-world scenarios where data acquisition is more likely to be affected by modality missing. For a given paired RGB image $\mathbf{D}_i^{\text{rgb}}$ and 3D data $\mathbf{D}_i^{\text{3D}}$, based on the data missing rules defined in Eq. (1), the input $\mathbf{X}_i$ to the model, which is generated by modality missing at the input level, is defined as:

$$\mathbf{X}_i = \langle \mathbf{D}_i^{\text{rgb}} \otimes \mathbf{M}_i^{\text{rgb}}, \mathbf{D}_i^{\text{3D}} \otimes \mathbf{M}_i^{\text{3D}} \rangle \tag{4.2}$$

Where element-wise multiplication is indicated by $\otimes$. Then, for multimodal representation, $\mathbf{X}_i$ is fed into the feature extractor Ω:

$$\langle \mathbf{F}_i^{\text{rgb}}, \mathbf{F}_i^{\text{3D}} \rangle = \Omega(\mathbf{X}_i). \tag{4.3}$$

Feature-Level Missing

Unlike modality missing at the input level, modality missing at the feature level is specifically designed for methods that are well-equipped with a feature alignment or pre-extraction module such as memory banks and feature galleries. In other words, as the feature extractor Ω is frozen, there is no unfair computational benefit for modality missing at the feature level. Thus, modality missing at the feature level is considered a **pseudomissing** approach, where the dataloader remains complete:

$$\mathbf{X}_i = \langle \mathbf{D}_i^{\text{rgb}}, \mathbf{D}_i^{\text{3D}} \rangle. \tag{4.4}$$

Feature extraction from modality-complete input $\mathbf{X}_i$ is defined as:

$$\langle \mathbf{F}_i^{\text{rgb}}, \mathbf{F}_i^{\text{3D}} \rangle = \Omega(\mathbf{X}_i). \tag{4.5}$$

The missing operation is then applied to the extracted feature maps:

$$\langle \tilde{\mathbf{F}}_i^{\text{rgb}}, \tilde{\mathbf{F}}_i^{\text{3D}} \rangle = \langle \mathbf{F}_i^{\text{rgb}} \otimes \mathbf{M}_i^{\text{rgb}}, \mathbf{F}_i^{\text{3D}} \otimes \mathbf{M}_i^{\text{3D}} \rangle. \tag{4.6}$$

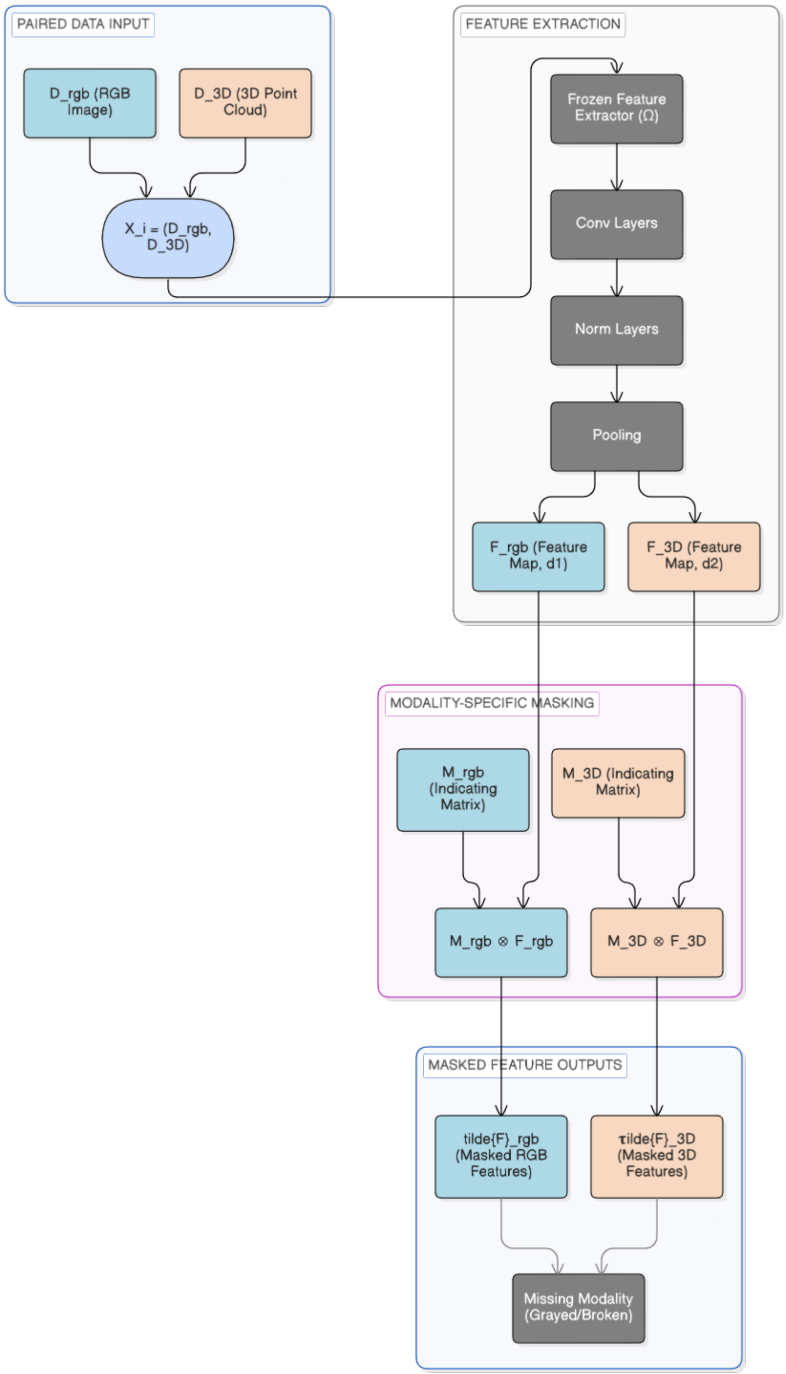

Figure 4.1: Architecture of the Feature-Level Pseudomissing process. Modalities are processed through a frozen encoder Ω to maintain consistent computational overhead, followed by element-wise multiplication ($\otimes$) with an indicating matrix M to simulate latent-space modality absence.

Rationale for Dual Missing Protocols. We mainly adopt the **input-level missing** (Eq. 2), because it more accurately represents real deployment scenarios where sensors malfunction before data reaches the model. This setting tests the model's capacity to extract reliable representations from incomplete inputs by forcing the feature extractor to interpret actual absence instead of zero-filled placeholders. The **feature-level missing** (Eq. 6) is also included for the purpose of comparison with memory bank-based methods like M3DM [28] and Shape-Guided [29], which involve pre-extracted feature maps for fusion. In this case, the baseline model and our method have access to the pre-computed feature galleries.

4.2 Model Specification

The proposed framework is an extension of the Contrastive Language-Image Pre-training (CLIP) [4] model for the specific application of industrial anomaly detection with multimodal inputs and missing modality scenarios. It comprises the Vision Transformers (ViT) [3] for vision encoding and the transformer-based text encoder for semantic alignment.

4.2.1 Model Architecture

The system's multi-encoder architecture includes:

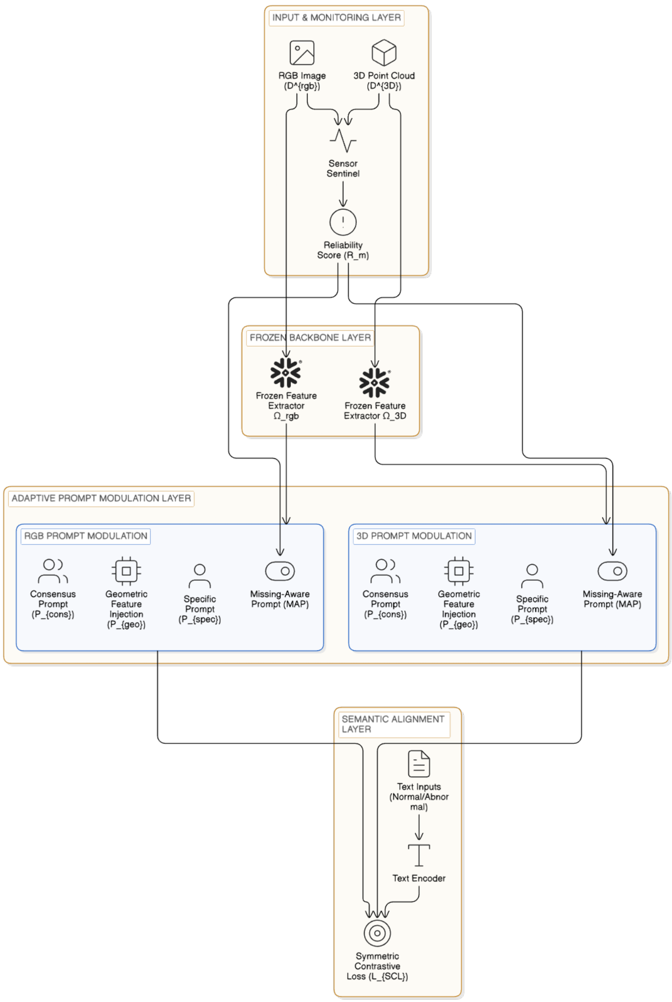

Figure 4.2: Proposed MISDD-MM architecture featuring frozen encoders (Ω), reliability-guided Sensor Sentinel (R_m), and geometric feature injection (P_{geo}) for resilient defect detection.

- **RGB Visual Encoder** (Ω_{rgb}): RGB images are processed by a Vision Transformer (ViT-B/16) with a constant self-attention mechanism. The definition of consistent self-attention is:

$$\text{Attention}(V, V, V) = \text{softmax}\left(\frac{VV^\top}{\sqrt{d_k}}\right)V,$$

  which improves defect localization by strengthening local semantic connections.

- **3D Visual Encoder** (Ω_{3D}): An RGB-like ViT-B/16 encoder is utilized for depth map inputs. PointMAE handles the 3D geometric data for point cloud inputs.

- **Text Encoder** (E_T): A transformer-based text encoder that is inherited from CLIP [4] generates semantic embeddings T_{n}, T_{an} from normal and abnormal text prompts, respectively.

4.2.2 Base Framework: Cross-Modal Prompt Learning

Three different kinds of learnable prompts are introduced and fed into the transformer layers to address dynamic modality missing. The formulation of the composite prompt mechanism is:

$$P = [P_{\text{cons}} \parallel P_{\text{spec}} \parallel P_{\text{miss}}]$$

where $\parallel$ denotes concatenation. The three prompt elements are:

Cross-Modal Consistency Prompt (P_{cons}) a randomly initialized learnable parameter matrix $P_{\text{cons}} \in \mathbb{R}^{l \times d}$, here l the prompt length and d the embedding dimension. This prompt establishes information consistency between RGB and 3D modalities across different input patterns, ensuring a unified semantic buffer area for surface defect localization.

Modality-Specific Prompt (P_{spec}) generated from the input features using self-attention and updated across layers, allowing the model to adapt its representations based on the available modality information. :

$$P_{\text{spec}} = \Pi_{i=1}^{h}\text{softmax}\left(\frac{W_h X_m \left(W_h X_m\right)^\top}{\sqrt{d_k}}\right) W_h X_m$$

Π denotes concatenation across h attention heads, W_h are projection matrices, X_m is the input from modality $m \in \{\text{RGB}, \text{3D}\}$, and d_k is the key dimension.

Missing-Aware Prompt (P_{miss}) attached to every transformer layer to fill in information gaps caused by missing modalities. In order to assist the model in adapting and prevent broken representations, MAP updates with the input features.

The prompts are injected into the early transformer layers (layers 0 to 6 by default) through multi-head attention followed by sequence concatenation and then sent through the standard attention and MLPs.

4.2.3 Symmetric Contrastive Learning

The SCL architecture is an adaptation of the dual-branch architecture, in which the RGB and 3D modalities are aligned independently with the text modality. In the case of the vision-language contrastive learning, the same architecture and mechanism of learning are shared by the two branches, which run in parallel and symmetrically with respect to the text modality. SCL uses two separate vision streams-RGB and 3D-PointMAE—each contrastively aligned with the same text prompts, in contrast to traditional CLIP [4], which uses a single vision encoder.

This creates symmetric learning objectives where both modalities independently learn to recognize defect semantics through shared textual anchors. Crucially, this architecture enables **single-modality inference**: when one sensor fails during deployment, the remaining modality can still perform defect detection by aligning its features with the learned text embeddings, without requiring cross-modal fusion.

Antithetical Text Prompts

The model generates paired text prompts to represent binary states:

- **Normal Prompt (T_n):** Descriptions of the object in a perfect state (e.g., "a photo of a [class]").
- **Abnormal Prompt (T_{an}):** Descriptions of the object with defects (e.g., "a photo of a [class] with [defect]").

Optimization Objective

The symmetric contrastive loss is defined as follows, pulling normal samples toward T_n and pushing them away from T_{an}:

$$\begin{aligned}\mathcal{L}_{\mathrm{SCL}} = \lambda_1 \big[d(F_{\mathrm{rgb}}, T_n) + d(F_{\mathrm{3D}}, T_n) \big] \\ + \lambda_2 \big[\max(0, m - d(F_{\mathrm{rgb}}, T_{an})) + \max(0, m - d(F_{\mathrm{3D}}, T_{an})) \big]\end{aligned} \tag{4.7}$$

Where:

- $\mathcal{L}(\cdot)$ denotes the L2 norm distance between two feature vectors.
- F_m represents the visual feature embedding from modality m, where $m \in \{RGB, 3D\}$.
- T_t represents the text feature embedding for state t, where $t \in \{n \text{ (normal)}, an \text{ (abnormal)}\}$.

In order to provide robust alignment in the shared embedding space, this formulation pushes RGB and 3D representations of normal samples away from the abnormal text embedding T^{an} and promotes them to cluster close to the normal text embedding T^n.

4.3 Data Collection

The study utilizes the **MVTec 3D-AD** dataset [30], widely recognized as the standard benchmark for multimodal industrial surface defect detection. This dataset is unique in its provision of high-resolution RGB images paired with high-precision 3D geometric data (point clouds) for various rigid and non-rigid industrial objects.

Table 4.2: Data Sources and Characteristics

Data Characteristic	**Description**
Dataset Name	MVTec 3D-AD
Modalities	High-resolution RGB Images + 3D Point Clouds (XYZ)
Total Categories	10
Defect Types	Scratches, cracks, dents, holes, contaminations, geometric deformations
Training Split	2,656 samples (Normal / Non-defective only)
Testing Split	1,197 samples (Mixture of Normal and Defective)
Missing Simulation	Dynamic masking with rate $\eta \in [0.0, 1.0]$ applied during training

4.3.1 Data Cleaning

To ensure the quality of the multimodal inputs certain cleaning techniques are used.

- **Structure Validation:** The existence of `train/good`, `test/good` and `test/defect` subdirectories for each of the ten categories is ensured by rigorous validation of the dataset structure.

- **3D Outlier Removal (Connected Components):** Background noise and floating anomalies are common in raw 3D scans. The point clouds are applied to a **Connected Components Cleaning** technique. This technique efficiently filters out isolated noise points by calculating the Euclidean distance between points to build clusters and keeping just the largest linked component, denoted as the principal object.

- **Pair Integrity:** RGB and 3D files are strictly synchronized by the loading pipeline. Any sample that does not have a matching pair because of corrupted files or missing indices is instantly removed from the training batch.

4.3.2 Data Transformation

The raw sensor data is normalized and transformed into tensors that are compatible with the pre-trained CLIP backbone and Point Transformer.

- **From Raw RGB to Normalized Tensors:** The dataset consists of high-resolution 8-bit color images. These are first resized to 224×224 pixels to match the CLIP backbone's positional embeddings [4]. The raw pixel values

$[0, 255]$ are then linearly scaled to $[0, 1]$ and normalized as per the OpenCLIP distribution: $[\mu = 0.4814, 0.4578, 0.4082], \sigma = [0.2686, 0.2613, 0.2757]$.

- **From Absolute Coordinates to Scale-Invariant Points:** The raw xyz coordinates, which are back-projected from the 16-bit depth maps, are absolute spatial positions in millimeters. To ensure that the model is able to learn the geometry as opposed to the absolute distance from the sensor, zero-mean centring and unit sphere scaling are employed. This maps the entire object into a localized coordinate space with its center at $(0, 0, 0)$ and radius 1.0.

- **Geometric Reduction via FPS:** The raw scans contain millions of raw xyz points, which need to be reduced down to $N = 4096$ points using the FPS method. This is a critical step in preserving the global geometry of the object while maintaining a uniform point density that is conducive to the Point Transformer encoder.

- **Semantic Tokenization:** The raw text prompts are tokenized into 77 tokens each by the CLIP BPE tokenizer, which produces a fixed-length sequence of token IDs from natural language strings.

4.3.3 Integrated Modalities in the Proposed Framework

The methodology allows for robust multimodal industrial surface defect detection, even in the case of missing or degraded sensor inputs, by combining **two sensor-collected modalities** with **one synthesized semantic modality**. The following is a definition of the three modalities:

1. **Visual Modality (RGB images)** captures surface appearance information such as texture, color, reflectance and fine visual features. Images are directly obtained from the RGB camera sensor employed in the industrial setting. RGB images are the primary sources of photometric information with high resolution.

2. **Geometric Modality** provides information regarding depth, surface topology, and surface curvature. 3D sensors such as structured light, laser triangulation, or time-of-flight cameras are employed to obtain this modality. This modality is particularly useful in detecting geometric anomalies such as dents, deformations and protrusions, which are not easily identifiable from RGB images.

3. **Semantic Modality** is made up of text data, which is automatically produced from a set of templates describing the defects and the classes. Examples of text data are "A scratched metallic surface", "A dented plastic surface", "A scratched bagel" and "Cracks on the ceramic coating". Text data acts as a semantic alignment anchor and is accessible even if one or both modalities are absent or corrupted.

The methodology leverages complementary information across photometric, geometric, and linguistic representations and increases robustness to sensor failures and modality dropouts by integrating raw sensor data (RGB + 3D) with a constructed and always-available semantic modality.

4.3.4 Data Reduction

Considering the computational cost of processing dense 3D scans, which is usually above 50,000 points per scan, dimensionality reduction is critical.

- **Farthest Point Sampling (FPS):** The system uses Farthest Point Sampling to reduce the dense point clouds to a set of representative $N = 4096$ points. FPS selects the point that is most distant from the set of previously chosen points. This process ensures the downsampled set maintains the same density and the global geometric structure of the object.

- **Image Resizing:** Resizing the RGB images to $\mathbf{224 \times 224}$ pixels using bicubic interpolation.

4.3.5 Summary of Preprocessed Data

Batches of synchronized tensors make up the final preprocessed dataset that is passed to the model.

Table 4.3: Final Preprocessed Data Tensor Specifications

Data Component	Tensor Shape	Data Type	Notes
RGB Image	$(B, 3, 224, 224)$	Float32	Normalized via OpenCLIP stats
Point Cloud	$(B, 4096, 3)$	Float32	XYZ coords, FPS sampled
Text Tokens	$(B, 77)$	Int64	Token IDs (max seq len 77)
Missing Mask	$(B, 2)$	Binary	Indicator for missing sensors

4.4 Implementation of Selected Design

The deep learning library used to implement the methodology presented here is the PyTorch library and the programming language used to do this is the Python programming language. The system's architecture is modular, divided into discrete parts for training, evaluation, model definition and data loading.

4.4.1 Implementation Environment

The implementation makes use of specialized software libraries to ensure efficiency.

4.4.2 Implementation Process

The workflow has been divided into three main phases as follows:

- **Model Initialization:** The script `MISDD_MM/model.py` initializes the ViT-L-14 CLIP backbone. To retain pre-trained knowledge, the visual encoders are frozen, whereas the PromptLearner and Missing-PromptLearner modules are initialized for learning the parameters P_{cons}, P_{spec}, and P_{miss}.

Table 4.4: Environment of Implementation

Component	**Specification**
Programming Language	Python 3.11
Deep Learning Framework	PyTorch 2.2.0 with CUDA 12.1
Key Libraries	OpenCLIP, Pointnet2_PyTorch, Open3D, NumPy, Scikit-learn
Hardware	NVIDIA GPU (Tested on CUDA 12.1 compatible devices)
Backbone Model	ViT-L/14 (CLIP)
Operating System	Linux (Ubuntu x86_64)

- **Training Loop:** The script `train_cls.py` follows this phase. Here, a dynamic missing mask is computed based on the missing rate η. The forward pass is performed by injecting learnable prompts into the Transformer layers and then computing the symmetric contrastive loss.

- **Optimization:** The model is optimized with the AdamW optimizer, which is standard for prompt learning. The learning rates of the prompt parameters are set for stable convergence without disturbing the frozen backbone.

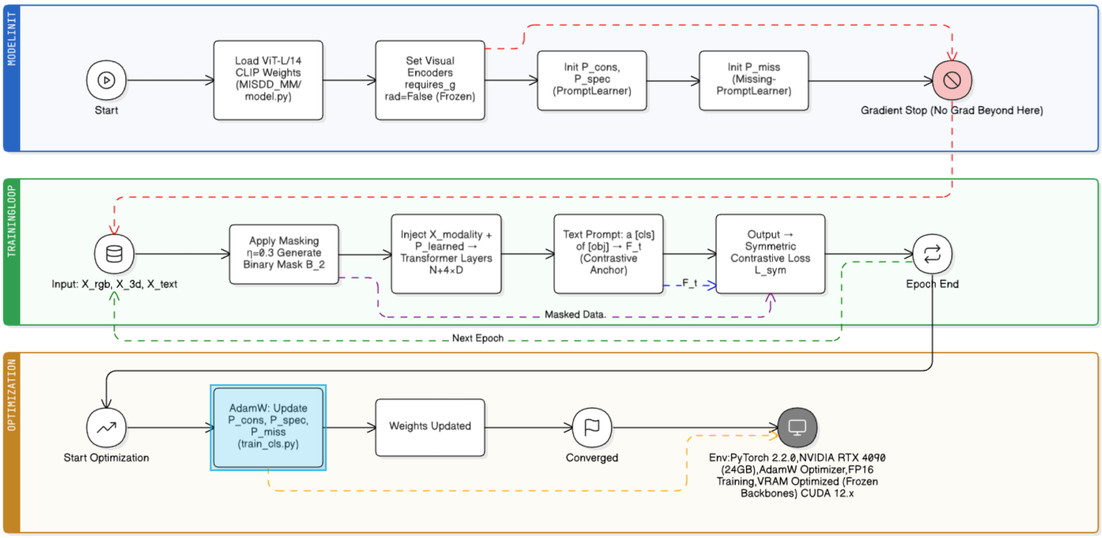

Figure 4.3: Workflow of the iterative implementation of the MISDD framework, showing the transition from the initialization of the model and the learning of the prompt parameters to the dynamic optimization of the missing mask.

4.4.3 Training Configuration

The model is trained with the following hyperparameters as defined in `params.py` and `train_cls.py`:

- **Batch Size:** 32 (default configuration)

- **Training Epochs:** 100

- **Image Resolution:** 224 $\times$ 224 pixels (matching standard CLIP ViT input)

- **Point Cloud Sampling:** 4096 points (via Farthest Point Sampling)
- **Prompt Configuration:** 4 context tokens, depth of 4 layers for `PromptLearner`
- **Missing Rate (η):** Variable (default 0.3 for training) to simulate sensor uncertainty

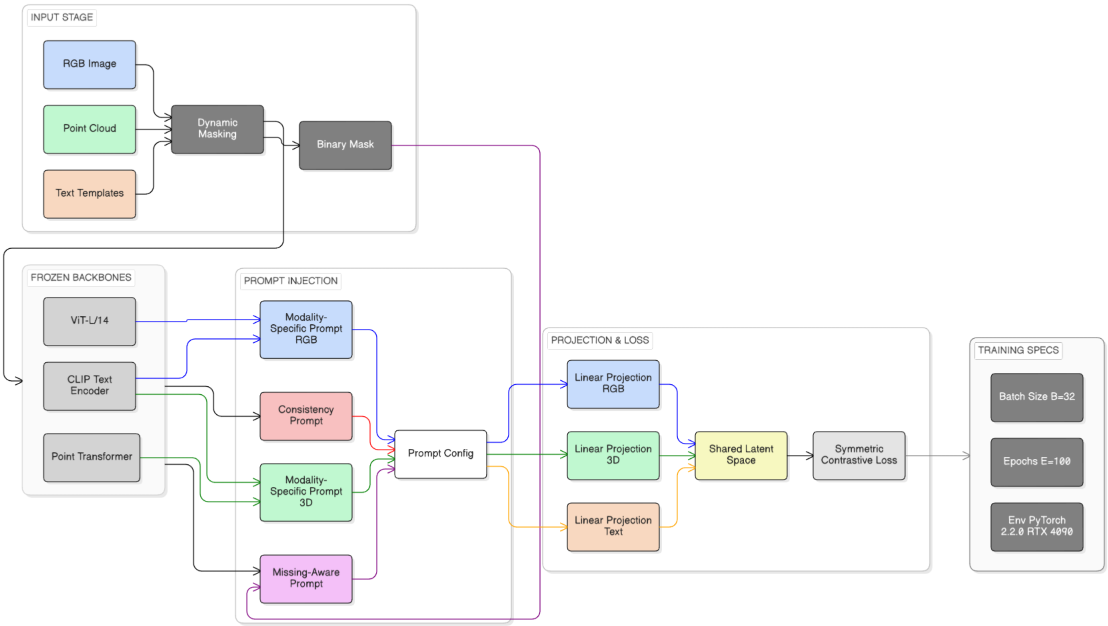

Figure 4.4: Configuration overview of the MISDD-MM training pipeline. Shown are the modality-specific sampling parameters, learnable prompt depths, and the symmetric contrastive alignment for resilient defect detection.

Finally, we proposed a robust approach for Multimodal Industrial Surface Defect Detection (MISDD) that is intended to function well in the event of unpredictable sensor availability. A generic **Missing Modality Configuration** to mimic sensor failures, **Cross-Modal Prompt Learning** to dynamically adapt feature extraction and **Symmetric Contrastive Learning** to match vision and geometric representations with semantic textual descriptions are all integrated into the suggested **MISDD-MM** framework [2].

The system achieves state-of-the-art resilience without the computational complexity of full fine-tuning by injecting learnable prompts ($P_{cons}, P_{spec}, P_{miss}$) into a frozen ViT-L/14 backbone. Using PyTorch 2.2.0, the implementation details verify that this concept is workable on contemporary hardware (NVIDIA RTX 4090), guaranteeing both a fast inference rate and geometric accuracy through Farthest Point Sampling.

4.4.4 Justification of Design Choices

The following justifies the choice of the dataset and model architecture:

MVTec 3D-AD as a Testbed for Missing-Modality Scenarios.

This work evaluates robustness to **sensor failures** during inference, not to actual data unpairing, using a dynamic missing-rate simulation (η). RGB and 3D inputs are first synchronized and spatially aligned in MVTec 3D-AD [30].

This carefully regulated procedure separates modality missing from confounding factors such as temporal disconnection or misalignment. Future extension will focus on:

- cross-dataset modality fusion,
- temporal unpairing (e.g., historical RGB + contemporary 3D),
- partially misaligned or inconsistent multimodal data streams.

The research focuses on the **missing-modality robustness** problem as defined in the MISDD-MM task [2]. The model develops modality-agnostic representations that are resilient in the event that one or both sensors are absent by using dynamic missing masks during training. This eliminates the need for the modalities to be truly unpaired or gathered from various sources.

CLIP and Prompt Learning for Semantic Bridging

The choice of the CLIP [4] backbone is crucial for addressing the missing modality problem. Most traditional fusion-based methods try to align the RGB features with 3D features. This is not feasible if one modality is missing.

- **Semantic Anchoring:** Our approach utilizes text data (defect descriptions) as a shared "anchor". This means that if the RGB and 3D data are missing (we have an image but no point cloud), the information can still be aligned to the text embedding.
- **Prompt-Based Alignment:** The Cross-Modal Consistency Prompt is designed to enhance the cross-modal representation learning by aligning the RGB and 3D embeddings to map to the same location in the shared latent region. Ensuring that the framework learns to represent the data in a unified modality-agnostic way.

4.4.5 Proposed Extensions and Future Work

While the MISDD-MM [2] approach is successful for addressing the *information vacancy* problem due to dynamic modality absence using missing-aware prompts and cross-modal prompt learning, there are limitations to the approach. First, the approach assumes that the modality information is either fully present or fully absent. This is not true for real-world scenarios where the quality of the data collected by the sensors can be degraded over time due to factors such as illumination changes, lens contamination, changes in surface reflectance, etc. Moreover, the approach is based on implicit feature learning using 3-D data. We propose two extensions, *Sensor Sentinel for predictive uncertainty estimation* and *Geometric Feature Injection* *for geometry-enhanced representation learning*, to further develop MISDD-MM [2] toward a robust multimodal industrial inspection system.

Sensor Sentinel: Predictive Uncertainty Estimation

In the current MISDD-MM framework [2], the missing modalities are simulated using a fixed missing probability η and the model adapts using missing-aware prompts when one modality is missing. However, this binary modeling does not take into account the reliability uncertainty of the sensor, which is common in industrial environments.

Proposed Design. In this research, we propose an additional module named *Sensor Sentinel*, which should be placed before the multimodal backbone. The Sensor Sentinel should be used for reliability estimation of each modality before feature extraction. In this case, instead of using binary modality indicators, we propose using a continuous modality reliability estimation $R_m \in [0, 1]$, for each modality $m \in \{\mathrm{RGB}, 3\mathrm{D}\}$.

Reliability Estimation. The reliability estimation score is derived using modality-specific statistical indicators. For the RGB modality, the indicators are signal-to-noise ratio (SNR), illumination variance, and texture entropy. For the 3-D modality, the indicators are point density variance, depth continuity, and surface smoothness. The reliability estimation is defined as:

$$R_m = f_m(\mathbf{S}_m), \tag{4.8}$$

where $f_m(\cdot)$ is a lightweight regression network and $\mathbf{S}_m$ represents a vector of sensor-quality statistics.

Integration with Prompt Learning. The estimated reliability score dynamically modulates the missing-aware prompt P^m_{miss} as

$$\tilde{P}^m_{\text{miss}} = R_m \cdot P^m_{\text{base}} + (1 - R_m) \cdot P^m_{\text{comp}}, \tag{4.9}$$

where P^m_{base} represents the standard modality-specific prompt and P^m_{comp} denotes the compensation prompt learned for missing modalities. This mechanism enables early and gradual activation of missing-modality compensation, allowing the model to adapt before a sensor fully fails.

Geometric Feature Injection for Robust 3-D Representation

The existing MISDD-MM framework [2] relies on the Point Transformer-based backbone to learn the geometric representation implicitly from the coordinates (x, y, z). Although the existing approach is effective in learning the 3D representation, the implicit learning of the 3D representation may face difficulties in handling subtle defects such as shallow dents or micro-deformations of the surface, especially when the RGB texture information is unavailable.

Proposed Design. We propose to inject the high-order geometric features into the 3D branch to learn the representation more effectively.

Geometric Feature Computation. To inject the geometric features into the 3-D branch, the surface normals $\mathbf{n}_i$ are computed. The mean curvature and Gaussian curvature are defined as:

$$H_i = \frac{1}{2}(k_1 + k_2), \tag{4.10}$$

$$K_i = k_1 \cdot k_2, \tag{4.11}$$

where k_1 and k_2 represent the principal curvatures at point i.

Geometric Prompt Injection. Introducing a geometric-specific prompt instead of directly concatenating these characteristics with raw point coordinates.

P_{geo}, initialized using curvature and normal statistics:

$$P_{\text{geo}} = g\big(\{\mathbf{n}_i, H_i, K_i\}\big), \tag{4.12}$$

where a learnable projection function is indicated by $g(\cdot)$. By injecting the geometric prompt into the 3D encoder's transformer layers, the model is able to maintain unambiguous topological cues even in situations where RGB is either missing or degraded.

4.4.6 Impact on Multimodal Representation Learning with Unpaired Data

The proposed extensions will directly benefit multimodal representation learning, especially for unpaired and unreliable data.

Soft-Unpaired Representation Learning: In conventional multimodal learning, data is either paired or unpaired based on the presence of modality. The Sensor Sentinel proposes the concept of *soft-unpaired data* where modalities may or may not be present but are considered unreliable due to low R_m.

Strengthened Single-Modality Representations: A significant challenge of multimodal learning with unpaired modalities is ensuring that any single modality representation is sufficiently informative. By augmenting the 3D representation with curvature-aware geometric prompts, the model will be able to generate complete defect representations even during 3D-only inference.

Future Plan:

Currently in the pre-thesis stage, we focus on implementing and benchmarking the MISDD-MM [2] baseline. Subsequently, we will integrate both the Sensor Sentinel and Geometric Feature Injection modules into the proposed architecture and extensively evaluate their effects on robustness with progressively degraded sensor information, performance during unpaired RGB-only and 3D-only scenarios and generalization with various industrial patterns of defects. The goal of these extensions is to make MISDD-MM [2] a robust, sensor-aware and geometry-informed multimodal defect detection platform framework.

4.4.7 Computational Complexity

To establish the feasibility of the proposed framework for deployment in an industrial scenario, we will analyze its computational complexity. We will measure the complexity of the proposed architecture by analyzing its trainable parameters, frozen parameters, floating point operations (FLOPs) and inference time. All experiments were performed utilizing an NVIDIA RTX 4090 GPU with an input size and batch size of 1 and an input size of 224x224 (RGB) + 224x224x3 (point cloud projection).

Table 4.5: Computational complexity breakdown of the proposed framework components.

Component	Parameters	FLOPs	Latency (ms)
ViT-L/14 Backbone	304M (frozen)	35.2G	12.3
Prompt Learner	0.5M	0.08G	0.4
Sensor Sentinel	8K	0.002G	0.1
Geometric Injection	12K	0.005G	0.3
Total	**304.52M**	**35.29G**	**13.1**

The computational expense ($\sim$99.8% of parameters and $\sim$99.7% of FLOPs) is dominated by the core ViT-L/14 backbone, which stays frozen during training and inference. Only 520,000 parameters (sim0.17% of total) and 0.087 GFLOPs (sim0.25% overhead) are contributed by the three suggested lightweight additions: Prompt Learner, Sensor Sentinel and Geometric Feature Injection.

The extra modules add less than 0.8 ms of inference time, according to latency measurements, making the overall end-to-end latency per sample 13.1 ms. This indicates *Sensor Sentinel* and *Geometric Injection* (P_{geo}) both are extremely deployment-friendly and impose a low computational burden, with a small increase ($< 1\%$) over the baseline ViT-L/14 model.

These results shows that the framework maintains near-real-time inference capability suitable for high-throughput industrial inspection systems while achieving substantial resistance to missing modalities and sensor degradation.

4.5 Conclusion

This research presents a comprehensive framework for multimodal industrial surface defect detection, which is capable of maintaining high levels of diagnostic accuracy despite varying levels of sensor reliability. Through the incorporation of a frozen ViT-L/14 CLIP backbone model, **Cross-Modal Prompt Learning** (CPL) and **Symmetric Contrastive Learning** (SCL), the research establishes text as a universal semantic bridge to align visual and geometric features, even when data is unpaired or unavailable. The proposed implementation utilizes high-resolution images from the MVTec 3D-AD dataset [30], which is optimized through Farthest Point Sampling (FPS) and coordinate normalization to ensure scale-invariant representations. To further refine the detection process, the proposed system utilizes a predictive **Sensor Sentinel** for real-time uncertainty estimation, as well as **Geometric**

Feature Injection to refine the fusion mechanism to prioritize high-order topological features such as surface normals and curvature, creating a geometry-aware detection system that is capable of maintaining high levels of detection accuracy even when the visual modality is degraded or unavailable.

Chapter 5

Result Analysis

5.1 Performance Evaluation

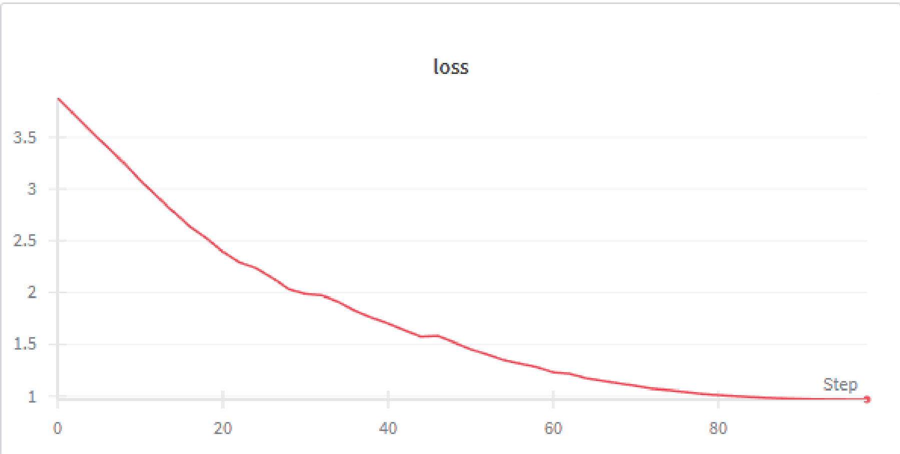

Figure 5.1: Convergence of the training loss for the proposed multimodal framework under uncertain sensor availability, showing the consistency and stability of the optimization process during training.

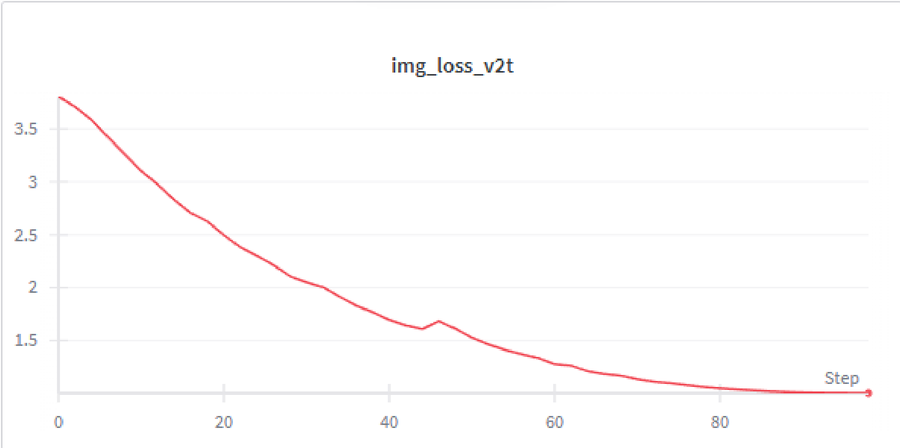

Figure 5.2: Convergence of the image-to-text contrastive loss $\mathcal{L}_{v2t}$ during training, showing the effectiveness of the image-text representation alignment process.

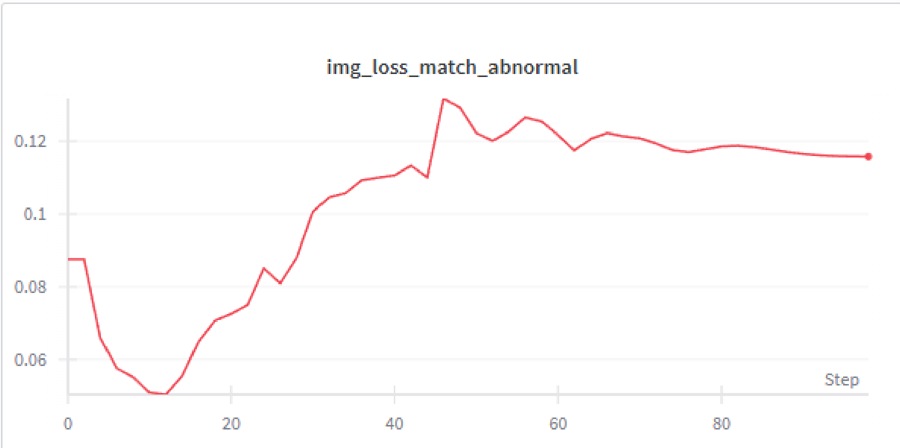

Figure 5.3: Training curve for the abnormal image matching loss, showing the progressive learning process for the discriminative abnormal image representations.

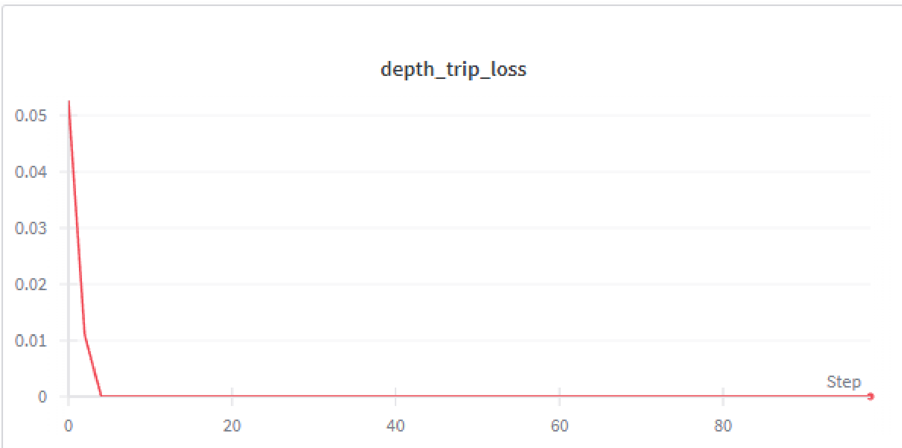

Figure 5.4: Convergence curve for the depth modality triplet loss, showing the effectiveness and rapidness of the metric learning process for the separation of image features using the depth modality.

Figure 5.5: Training curve for the image-level triplet loss, showing the effectiveness and rapidness of the metric learning process for the separation of normal and abnormal image samples in the image embedding space, validating the effectiveness and stability of the proposed symmetric contrastive learning strategy.

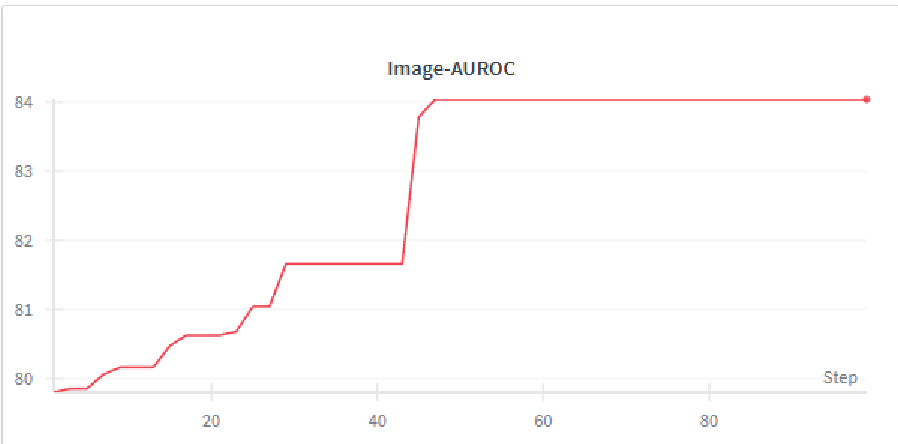

Figure 5.6: Image-level AUROC curve for the image-level abnormality classification process, showing the effectiveness of the proposed image-level multimodal fusion framework under uncertain sensor availability for image-level defect discrimination..

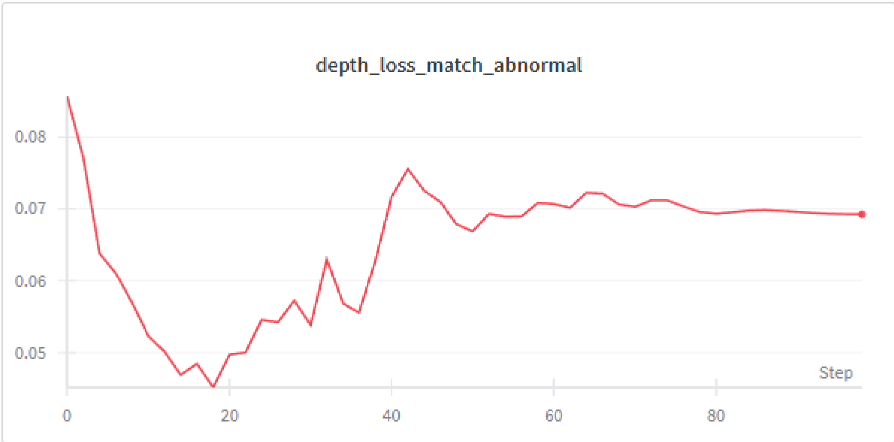

Figure 5.7: Training curve of the depth matching loss for abnormal samples, showing the effectiveness and stability of the image classification process under uncertain sensor availability due to the information vacancy caused by the dynamic missing of image modalities through the process of cross-modal prompt learning.

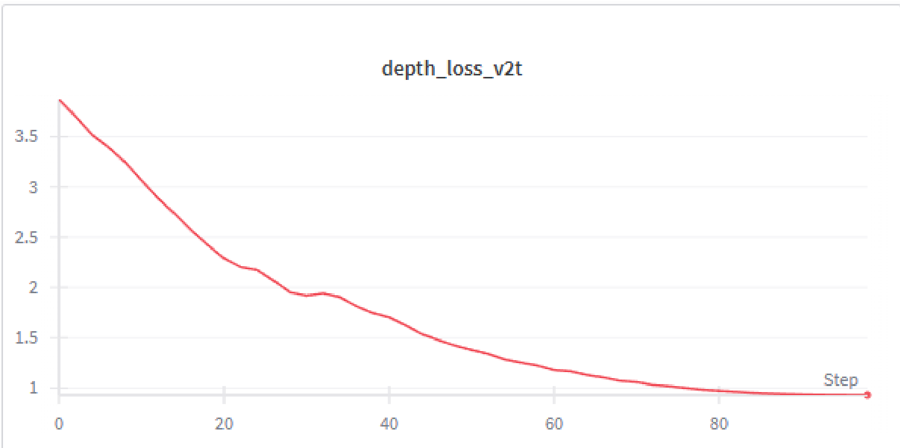

Figure 5.8: Depth reconstruction loss during training. Demonstrating the effectiveness of the image classification process by the model for the utilization of 3D structural information for the localization process..

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